METHODS DEVELOPMENT NOTE / REVISED ROUGH DRAFT

Extending the amyloid mixture-cure ODE model to longitudinal tau PET

Three susceptibility states, a departure-age copula, and amyloid-dependent tau progression

Status. This revision replaces the previous speed copula with a dynamic amyloid-to-tau rate multiplier. It preserves the three-state mixture, separate biomarker measurement models, and joint Bayesian inference. It is a mathematical and implementation proposal; no joint fit, recovery result, or treatment effect is claimed. The accompanying code is editable LaTeX, not a completed R implementation.

Agreed structure. Keep states 00, 10, and 11; connect departure ages with one Gaussian copula; let current latent amyloid *burden* modify tau velocity after tau departure. Initially make the two residual individual speeds independent, conditional on covariates and population parameters. Keep individual tau speed to represent heterogeneity left after the coupling.

1 Observed data and notation

Let $i = 1, \ldots, N$ index subjects, $k \in \{A, T\}$ index amyloid and tau, and $j = 1, \ldots, n_{ki}$ index scans of biomarker k. Let a_{ij}^k be calendar age in years and y_{ij}^k the observed PET value. Define

$$D_i^k = \{(a_{ij}^k, y_{ij}^k) : j = 1, \ldots, n_{ki}\}, \qquad D = \{D_i^A, D_i^T\}_{i=1}^N.$$

Uppercase Y_{ij}^k denotes a random scan value. Ages, visit schedules, and covariates are conditioned on. Scans need not be contemporaneous. This draft uses one scalar tau summary with a specified region and tracer.

Symbol	Definition
$Z_{ki} \in \{0, 1\}$	Eventual susceptibility to accumulation of biomarker k; not current positivity.
$S_i = (Z_{Ai}, Z_{Ti})$	Susceptibility state; permitted values $00, 10, 11$.
α_{ki}	Departure age, in years, for an accumulator.
δ_{ki}	Dimensionless individual log-speed; $e^{\delta_{ki}}$ is its multiplier.
$b_{ki},\ \mu_{bk}$	Constant non-accumulator level; common accumulator launch level.
η_i	Latent vector $(\alpha_{Ai}, \alpha_{Ti}, \delta_{Ai}, \delta_{Ti}, b_{Ai}, b_{Ti})$.
Θ	All unknown population coefficients, scales, smoothing parameters, copula correlation, and coupling strength, defined below.

Reading guide. Sections 2–6 define the generative model and posterior. Sections 7–8 derive the revised sampling coordinates. Sections 9–12 give priors, code changes, and tests. Amber boxes identify assumptions or checks that must be addressed before scientific reporting.

2 Three states and a proper common latent distribution

Impose $Z_{Ti} \leq Z_{Ai}$. Let w_i and v_i be known susceptibility covariate vectors, including intercepts, and let ϕ_A and $\phi_{T|A}$ be their regression coefficients. Define $\text{expit}(x) = (1 + e^{-x})^{-1}$ and

$$\pi_i = \text{expit}(w_i^\top \phi_A), \qquad q_i = \text{expit}(v_i^\top \phi_{T|A}).$$

Here $\pi_i = P(Z_{Ai} = 1 \mid \Theta)$ and $q_i = P(Z_{Ti} = 1 \mid Z_{Ai} = 1, \Theta)$, with covariates suppressed in probability notation. Let $w_{si} = P(S_i = s \mid \Theta)$ for state s. Then

$$w_{00,i} = 1 - \pi_i, \quad w_{10,i} = \pi_i(1 - q_i), \quad w_{11,i} = \pi_i q_i. \tag{1}$$

These sum to one. State 10 means amyloid accumulation without eventual tau accumulation. Neither departure-age ordering nor positivity-age ordering follows from this restriction.

Non-accumulator levels. Let c_k be the biomarker-specific positivity threshold and $\sigma_{bk} > 0$ its non-accumulator scale. Let φ and Φ denote the standard-normal density and CDF. For $\mu_{bk} < c_k$, use the normalized density

$$f_{bk}(b) = \frac{\varphi((b - \mu_{bk})/\sigma_{bk})}{\sigma_{bk}\Phi((c_k - \mu_{bk})/\sigma_{bk})} \mathbf{1}\{b < c_k\}. \tag{2}$$

The normal location μ_{bk} is the launch level and *untruncated* normal mean, not the mean after truncation. The levels b_{Ai} and b_{Ti} are initially independent. Measurement error can put a scan above c_k even in this branch.

Common augmentation. Let $f_{\alpha i}$ be the joint departure-age density in Section 3 and g_{ki} the log-speed density in Section 4. Draw

$$f_i(\eta_i \mid \Theta) = f_{\alpha i}(\alpha_{Ai}, \alpha_{Ti}) g_{Ai}(\delta_{Ai}) g_{Ti}(\delta_{Ti}) f_{bA}(b_{Ai}) f_{bT}(b_{Ti}), \tag{3}$$

independently of S_i given Θ and covariates. Subjects are independent conditional on Θ. This is a deliberate generative assumption, not a consequence of the mixture notation.

Every subject retains the same continuous coordinates. Inactive departure ages, speeds, and flat levels are auxiliary variables with proper distributions. In state 10, tau departure and speed integrate out; in state 00, both accumulator histories integrate out. The amyloid history marginal is the same in states 10 and 11. The joint departure distribution has a biological paired-history interpretation in state 11 only.

TEST: susceptibility assumptions and limited follow-up

The no-tau-only restriction is a scientific simplification, particularly consequential for medial temporal tau. A negative amyloid scan does not establish lifetime non-susceptibility. Assess the restriction for the chosen tau summary and compare with a four-state sensitivity model if appropriate. Different observed positivity fractions, or different lower bounds on lifetime susceptibility, do not establish unequal lifetime susceptibilities. Late departure and non-accumulation can remain difficult to distinguish.

3 Departure-age marginals and their Gaussian copula

Let x_{ki} be a known departure-age covariate vector including an intercept, γ_k its coefficient vector, and $\sigma_{\alpha k} > 0$ a shape-control parameter. Let $a_{\min,k} < a_{\max,k}$ be fixed departure-age bounds. Set

$$\lambda_{ki} = (1 \text{ year}) \exp(x_{ki}^\top \gamma_k), \qquad \kappa_k = 1/\sigma_{\alpha k}, \qquad C_{ki} = 1 - \exp\left[-\left(\frac{a_{\max,k} - a_{\min,k}}{\lambda_{ki}}\right)^{\kappa_k}\right].$$

The one-year reference fixes units inside the log-scale regression. For $a_{\min,k} < a < a_{\max,k}$, the shifted, upper-truncated Weibull CDF and density are

$$F_{ki}(a) = \frac{1 - \exp[-((a - a_{\min,k})/\lambda_{ki})^{\kappa_k}]}{C_{ki}}, \tag{4}$$

$$f_{ki}(a) = \frac{\kappa_k}{\lambda_{ki} C_{ki}} \left(\frac{a - a_{\min,k}}{\lambda_{ki}}\right)^{\kappa_k - 1} \exp\left[-\left(\frac{a - a_{\min,k}}{\lambda_{ki}}\right)^{\kappa_k}\right]. \tag{5}$$

Extend F_{ki} by zero and one outside its bounds; $f_{ki} = 0$ outside. Truncation changes the distribution: it is not clipping an untruncated draw. The coefficients affect the underlying Weibull scale; with a fixed upper truncation, $e^{\gamma_{kq}}$ is not generally an exact ratio of observed truncated quantiles.

Constructing the dependence. Let V_{Ai}, V_{Ti} be latent standard-normal scores and $\rho_\alpha \in (-1, 1)$ their correlation. Let U_{ki} be their uniform percentile transforms. Generate

$$\begin{pmatrix} V_{Ai} \\ V_{Ti} \end{pmatrix} \sim \mathcal{N}_2\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho_\alpha \\ \rho_\alpha & 1 \end{pmatrix}\right), \quad U_{ki} = \Phi(V_{ki}), \quad \alpha_{ki} = F_{ki}^{-1}(U_{ki}). \tag{6}$$

The normal scores and percentiles are dimensionless; the final quantiles are ages in years. Explicitly, for $0 < u < 1$,

$$F_{ki}^{-1}(u) = a_{\min,k} + \lambda_{ki}[-\log(1 - uC_{ki})]^{1/\kappa_k}.$$

Let $\varphi_2(\cdot, \cdot; \rho)$ denote the bivariate standard-normal density with correlation ρ. For probability arguments $p_1, p_2 \in (0, 1)$, define

$$c_\rho(p_1, p_2) = \frac{\varphi_2(\Phi^{-1}(p_1), \Phi^{-1}(p_2); \rho)}{\varphi(\Phi^{-1}(p_1))\varphi(\Phi^{-1}(p_2))}.$$

For generic departure ages a_A, a_T, the joint density is

$$f_{\alpha i}(a_A, a_T) = c_{\rho_\alpha}(F_{Ai}(a_A), F_{Ti}(a_T)) f_{Ai}(a_A) f_{Ti}(a_T). \tag{7}$$

This preserves both marginals exactly. Positive ρ_α associates early-with-early and late-with-late departures relative to covariates. It is not generally the Pearson correlation of calendar ages. Independence occurs at $\rho_\alpha = 0$; strict temporal ordering is not imposed.

4 Independent residual speeds, rate curves, and domain

Let u_{ki} be a centered speed covariate vector *without an intercept*, ψ_k its coefficient vector, and $\sigma_{\delta k} > 0$ a residual scale. Let $c_z = 4$ be an initial fixed bound. Define

$$h(z) = \frac{\varphi(z)}{\Phi(c_z) - \Phi(-c_z)} \mathbf{1}\{-c_z < z < c_z\}, \qquad z_{Ai}, z_{Ti} \overset{\text{ind}}{\sim} h, \qquad \delta_{ki} = u_{ki}^\top \psi_k + \sigma_{\delta k} z_{ki}.$$

Thus, for a generic log-speed d,

$$g_{ki}(d) = \sigma_{\delta k}^{-1} h\left(\frac{d - u_{ki}^\top \psi_k}{\sigma_{\delta k}}\right). \tag{8}$$

There is **no speed copula and no** ρ_δ. Conditional on Θ and covariates, the departure-age pair is independent of both residual speeds. These are latent-distribution assumptions; the data can induce posterior dependence among timing and speed parameters. Total tau velocity will depend on the amyloid history even with independent residual speeds.

For each biomarker, let $B_k(y)$ and $V_k(y)$ be fixed spline basis vectors of lengths K_k and Q_k; let θ_k and ν_k be their coefficients. Define

$$r_k(y) = \exp\{B_k(y)^\top \theta_k\}, \qquad \omega_k(y) = \exp\{V_k(y)^\top \nu_k\}. \tag{9}$$

The numerical rate r_k is in biomarker units per year; ω_k is a measurement scale in biomarker units. Logarithms of these numerical quantities use fixed reporting units. Let $r_{\min,k} > 0$ be a fixed rate floor, and write $\bar{r}_k(y) = \max\{r_k(y), r_{\min,k}\}$. All trajectory equations below use $\bar{r}_k$, consistently with the clock.

Define the strictly increasing population clock and its inverse by

$$G_k(y) = \int_{c_k}^{y} \frac{dv}{\bar{r}_k(v)}, \qquad G_k^{-1} = \text{inverse of } G_k. \tag{10}$$

The integration variable v is a biomarker level. G_k has units of years and is anchored at the positivity threshold. Each biomarker retains its own rates, noise scales, launch level, threshold, and spline bases.

Specify tails rather than imposing an accidental prior. Let $[\ell_k, h_k]$ be a fixed interval containing the spline knots and all plausible launch levels. As an explicit initial convention, extend each log-rate and log-noise function constantly beyond the corresponding boundary. The basis functions in (9) include this extension. For finite coefficients, rates are then positive and bounded, clocks are defined on $\mathbb{R}$ with inverses on $\mathbb{R}$, and trajectories cannot explode in finite time. Clock tails are linear and can be evaluated analytically. This also defines measurement scales for the unbounded lower tail of b_{ki}.

TEST: numerical and tail assumptions

Tail continuation is a modeling choice, not saturation or a biological finding. Report how often fitted trajectories use it; vary the boundaries and continuation. Refine quadrature and interpolation, reduce the rate floor, and confirm posterior quantities are stable. A finite lookup table must not silently truncate the latent distribution or clamp physical predictions at its edge.

5 Coupled trajectories and the revised tau clock

Let $(x)_+ = \max(x, 0)$. Define the *candidate accumulator* amyloid path for every subject by

$$A_i^1(a) = G_A^{-1}\left[G_A(\mu_{bA}) + e^{\delta_{Ai}}(a - \alpha_{Ai})_+\right]. \tag{11}$$

It is physical in states 10 and 11, and auxiliary in state 00.

A parsimonious initial coupling. Let $d_A^{\text{coup}} > 0$ be a fixed transition width in amyloid units, and $\beta \geq 0$ an unknown dimensionless coupling strength. For an amyloid level y, define

$$H_A(y) = \operatorname{expit}\left(\frac{y - c_A}{d_A^{\text{coup}}}\right), \qquad s(y) = \beta\{H_A(y) - H_A(c_A)\}, \qquad m_i(a) = e^{s(A_i^1(a))}. \tag{12}$$

Thus $s(c_A) = 0$, $m_i(a) > 0$, and $m_i = 1$ at the reference amyloid burden. The multiplier is bounded between $e^{-\beta/2}$ and $e^{\beta/2}$ and increases with burden. A high, stable amyloid level can sustain a high multiplier; amyloid *velocity* does not enter. Setting $\beta = 0$ removes coupling. The sigmoid shape and width are provisional assumptions, not literature-established constants. A flexible centered coupling spline is a later sensitivity model.

Let $T_i^1(a)$ be the candidate tau accumulator path. It equals μ_{bT} for $a \leq \alpha_{Ti}$. After departure,

$$\frac{dT_i^1(a)}{da} = e^{\delta_{Ti}} m_i(a) \bar{r}_T(T_i^1(a)), \qquad T_i^1(\alpha_{Ti}) = \mu_{bT}. \tag{13}$$

Define its accumulated exposure clock, in years, by

$$W_i(a; \alpha_{Ti}) = \begin{cases} 0, & a \leq \alpha_{Ti}, \\ \displaystyle\int_{\alpha_{Ti}}^{a} m_i(u)\, du, & a > \alpha_{Ti}, \end{cases}$$

where u is calendar age. The chain rule gives $dG_T(T_i^1(a))/da = e^{\delta_{Ti}} m_i(a)$ after departure, hence

$$T_i^1(a) = G_T^{-1}\left[G_T(\mu_{bT}) + e^{\delta_{Ti}} W_i(a; \alpha_{Ti})\right]. \tag{14}$$

The population inverse clock still works. The new computation is a *subject-specific one-dimensional age integral*; simply substituting the current multiplier into $(a - \alpha_{Ti})_+$ is incorrect when amyloid varies. The amyloid equation has no tau feedback.

TEST: interpretation of the coupling

This coupling changes progression *after* tau departure; it does not cause or forbid departure. It assumes the same proportional amyloid effect at each tau burden, while $\bar{r}_T$ controls tau-level dependence. Test sensitivity to the fixed width, monotonicity, saturation, and stage dependence. An observational estimate of s is not automatically an amyloid-lowering treatment effect.

6 Observation likelihood, mixture, and posterior

For state $s \in \{00, 10, 11\}$, let $M_i^{k,s}(a)$ be the latent mean used for biomarker k:

s	$M_i^{A,s}(a)$	$M_i^{T,s}(a)$
00	b_{Ai}	b_{Ti}
10	$A_i^1(a)$	b_{Ti}
11	$A_i^1(a)$	$T_i^1(a)$

Tau coupling uses the amyloid *accumulator* path only in state 11. There is no tau-only branch requiring coupling to an amyloid flat path.

Let d_k be fixed Student-t degrees of freedom, initially $d_A = d_T = 4$, and let $t_d(y; m, \omega)$ be its density at y with location m and scale $\omega > 0$. Assume

$$Y_{ij}^k \mid S_i = s, \eta_i, \Theta \sim t_{d_k}\Big(M_i^{k,s}(a_{ij}^k), \omega_k(M_i^{k,s}(a_{ij}^k))\Big),$$

independently across biomarkers and visits conditional on these quantities. The conditional variance is $d_k\omega_k^2/(d_k - 2)$ for $d_k > 2$. Shared latent histories do not require correlated measurement errors.

Define L_{Ai}^0, L_{Ai}^1 as the products of amyloid scan densities using b_{Ai}, A_i^1, respectively; L_{Ti}^0 uses b_{Ti}. To emphasize its additional dependence, define

$$L_{Ti}^{11}(\eta_i, \Theta) = \prod_{j=1}^{n_{Ti}} t_{d_T}\Big(y_{ij}^T; T_i^1(a_{ij}^T), \omega_T(T_i^1(a_{ij}^T))\Big).$$

An empty product equals one. The state-marginalized conditional likelihood is

$$\mathcal{L}_i(\eta_i, \Theta) = (1 - \pi_i)L_{Ai}^0 L_{Ti}^0 + \pi_i(1 - q_i)L_{Ai}^1 L_{Ti}^0 + \pi_i q_i L_{Ai}^1 L_{Ti}^{11}. \tag{15}$$

All factors are conditional on the *same* proposed latent vector. In particular, L_{Ti}^{11} depends on amyloid timing, speed, rate curve, launch level, and coupling through A_i^1. This is not a product of separately marginalized biomarker likelihoods.

Let $p(\Theta)$ be a proper population prior. Using the latent density in (3),

$$p(\Theta, \eta_{1:N} \mid D) \propto p(\Theta) \prod_{i=1}^{N} \mathcal{L}_i(\eta_i, \Theta) f_i(\eta_i \mid \Theta), \tag{16}$$

$$p(D \mid \Theta) = \prod_{i=1}^{N} \int \mathcal{L}_i(\eta_i, \Theta) f_i(\eta_i \mid \Theta)\, d\eta_i. \tag{17}$$

MCMC samples continuous histories rather than numerically evaluating each high-dimensional integral. The three states are summed out exactly. The state-independent augmentation in Section 2 is essential to factoring f_i outside the state sum. State-specific latent distributions would need their respective densities inside that sum.

Missing scans. Empty products assume ignorable missingness/visit scheduling conditional on modeled information. Subjects with no tau scans can inform amyloid parameters; they do not directly identify tau susceptibility or timing dependence through nonexistent tau observations. After integrating unused tau coordinates, their likelihood reduces to the amyloid-only mixture.

7 Revised location coordinates and the correct Jacobian

For each biomarker let $\widetilde{a}_{ki}$ be its mean scan age, or a fixed reference age if unobserved. Let $\widetilde{x}_{ki}$ be a sampling location coordinate. Extend each population clock below its launch level:

$$G_{k,\mathrm{ext}}(x) = \begin{cases} G_k(x), & x \geq \mu_{bk}, \\ G_k(\mu_{bk}) + (x - \mu_{bk})/\bar{r}_k(\mu_{bk}), & x < \mu_{bk}. \end{cases}$$

Write $D_k(x) = G'_{k,\mathrm{ext}}(x)$, equal to $1/\bar{r}_k(x)$ above the launch level and $1/\bar{r}_k(\mu_{bk})$ below it. Define $\Delta G_{ki} = G_{k,\mathrm{ext}}(\widetilde{x}_{ki}) - G_k(\mu_{bk})$.

Amyloid transformation remains unchanged:

$$\alpha_{Ai} = \widetilde{a}_{Ai} - e^{-\delta_{Ai}} \Delta G_{Ai}, \qquad J_{Ai} = \left| \frac{\partial \alpha_{Ai}}{\partial \widetilde{x}_{Ai}} \right| = e^{-\delta_{Ai}} D_A(\widetilde{x}_{Ai}). \tag{18}$$

Build A_i^1 from this age. Define a signed cumulative exposure, anchored at the tau reference age,

$$C_i(a) = \int_{\widetilde{a}_{Ti}}^{a} m_i(u)\, du. \tag{19}$$

The integral is negative before the reference age. Since $C_i'(a) = m_i(a) > 0$, it has an inverse. Bounded positive coupling makes its range all of $\mathbb{R}$ under the stated tail convention.

Tau transformation must change:

$$\alpha_{Ti} = C_i^{-1}\left(-e^{-\delta_{Ti}} \Delta G_{Ti}\right), \qquad J_{Ti} = \frac{e^{-\delta_{Ti}} D_T(\widetilde{x}_{Ti})}{m_i(\alpha_{Ti})}. \tag{20}$$

To see the new denominator, differentiate $C_i(\alpha_{Ti}) = -e^{-\delta_{Ti}} \Delta G_{Ti}$ with respect to $\widetilde{x}_{Ti}$ holding the amyloid path and speeds fixed. Then $m_i(\alpha_{Ti})\, \partial \alpha_{Ti}/\partial \widetilde{x}_{Ti} = -e^{-\delta_{Ti}} D_T(\widetilde{x}_{Ti})$.

The joint transformation is *triangular*, not independent: α_{Ai} depends on $\widetilde{x}_{Ai}$ only, while α_{Ti} generally depends on both locations. At fixed speeds and Θ,

$$\frac{\partial(\alpha_{Ai}, \alpha_{Ti})}{\partial(\widetilde{x}_{Ai}, \widetilde{x}_{Ti})} = \begin{pmatrix} -J_{Ai} & 0 \\ * & -J_{Ti} \end{pmatrix}, \qquad p(\widetilde{x}_{Ai}, \widetilde{x}_{Ti} \mid \delta_{Ai}, \delta_{Ti}, \Theta) = f_{\alpha i}(\alpha_{Ai}, \alpha_{Ti}) J_{Ai} J_{Ti}.$$

The asterisk denotes a generally nonzero cross-derivative; it does not enter the determinant.

REQUIRED CHANGE: do not reuse the old tau density unchanged

Evaluate the Weibull marginals and timing copula at the newly implied ages and include $-\log m_i(\alpha_{Ti})$ in the tau log-Jacobian. Replacing only the tau trajectory while retaining the old departure map targets the wrong model. Do not add separate age priors on top of this induced density.

8 Coordinate support, posterior evaluation, and state summaries

The tau trajectory in location coordinates has a useful cancellation:

$$T_i^1(a) = G_T^{-1}\Big[\max\{G_T(\mu_{bT}),\ G_{T,\mathrm{ext}}(\widetilde{x}_{Ti}) + e^{\delta_{Ti}} C_i(a)\}\Big]. \tag{21}$$

At $a = \widetilde{a}_{Ti}$, a location above the launch level is the physical candidate level. A location below it encodes *future departure*; the physical path remains at μ_{bT} until departure. There is no added atom in the sampling coordinate and no free prior on $\widetilde{x}_{ki}$.

Auxiliary paths must remain common across states. Compute C_i and the coordinate transformation using A_i^1 even when evaluating state 00 or 10. There the unused tau path is auxiliary, not a physical accumulating trajectory. Switching to the flat amyloid path in the coordinate prior for state 00 would destroy the single common transformed density used here.

Let $\xi_i = (\widetilde{x}_{Ai}, z_{Ai}, b_{Ai}, \widetilde{x}_{Ti}, z_{Ti}, b_{Ti})$ be the sampled subject vector. With implied ages and speeds substituted, its posterior is

$$p(\Theta, \xi_{1:N} \mid D) \propto p(\Theta) \prod_i \mathcal{L}_i\, f_{\alpha i}(\alpha_{Ai}, \alpha_{Ti}) J_{Ai} J_{Ti}\, h(z_{Ai}) h(z_{Ti}) f_{bA}(b_{Ai}) f_{bT}(b_{Ti}). \tag{22}$$

No $1/\sigma_{\delta k}$ factors belong in a prior directly on z_{ki}; they are already accounted for by changing from δ to z. When implementing with numerical clocks, Jacobians must differentiate the *actual interpolation used*. For a piecewise-linear G_k or C_i, use its segment slope rather than an independently interpolated reciprocal rate or multiplier that is not its derivative.

Support. Check the implied ages against their stated bounds. Since C_i is increasing, tau feasibility is equivalent to

$$C_i(a_{\min,T}) < -e^{-\delta_{Ti}} \Delta G_{Ti} < C_i(a_{\max,T}).$$

Endpoints have zero probability. Do not clip the root, the CDF inputs, or a below-launch location to create feasibility. Numerical grids must represent the specified model, including tails. Rejecting proposals only because an auxiliary path leaves a finite table would introduce unintended parameter-dependent support.

Posterior state probabilities. Let $B_{00,i} = L_{Ai}^0 L_{Ti}^0$, $B_{10,i} = L_{Ai}^1 L_{Ti}^0$, and $B_{11,i} = L_{Ai}^1 L_{Ti}^{11}$. Define

$$\ell_{si} = \log w_{si} + \log B_{si}, \qquad r_{si} = \exp\{\ell_{si} - \mathrm{logsumexp}_{t\in\{00,10,11\}}\, \ell_{ti}\}.$$

Here logsumexp is a stable log-sum-exp. Each draw satisfies $\sum_s r_{si} = 1$; averaging gives posterior class probabilities. With $\mathbb{E}$ denoting posterior expectation, a biological tau-departure summary is

$$\mathbb{E}(\alpha_{Ti} \mid Z_{Ti} = 1, D) = \frac{\mathbb{E}(r_{11,i}\alpha_{Ti} \mid D)}{\mathbb{E}(r_{11,i} \mid D)}.$$

Use analogous weighted distributions for intervals. Unconditional auxiliary-age summaries are not biological estimates for non-accumulators.

9 Priors, scale constraints, and scientific interpretation

Population prior. Use proper priors for every free component of Θ. Retain the supplied amyloid prior families for an initial comparison, but calibrate tau priors to its units and region. Fixed settings include all thresholds, age bounds, knots, rate floors, c_z, coupling width, and prior hyperparameters. They must be frozen before a production fit or explicitly modeled.

Component	Proposed starting specification
Timing correlation	Let $U_\rho = (\rho_\alpha + 1)/2 \sim \text{Beta}(2, 2)$. This permits either sign and mildly favors zero. Compare a uniform correlation prior.
Coupling strength	$\beta \sim \mathcal{N}(0, 1)$ truncated to $\beta \geq 0$, provisionally. The maximum/minimum multiplier ratio is e^β; use that ratio for prior calibration. A no-coupling model fixes $\beta = 0$ separately.
Departure regressions	Proper normal priors for γ_k; positive proper priors for $\sigma_{\alpha k}$, initially the current bounded Gamma family. Check implied ages after truncation.
Speed regressions	No intercept in $u_{ki}^\top \psi_k$. Proper normal coefficient priors and bounded half-normal $\sigma_{\delta k}$ priors; independent truncated-standard-normal residuals.
Susceptibility	Proper normal priors for $\phi_A, \phi_{T\|A}$. Calibrate the latter through the implied q_i and state frequencies rather than copying the amyloid intercept.
Launch and flat levels	Proper normal prior on μ_{bk} truncated to $\ell_k < \mu_{bk} < c_k$; proper positive prior for σ_{bk}; normalized f_{bk} from (2).
Measurement curve	Proper initial normal prior and bounded normal random-walk increments for ν_k, as in the current family, calibrated separately for tau.

For rate smoothing, let $\zeta_k > 0$ be a global precision and $\lambda_{kj}^\theta > 0$ a local variance multiplier for adjacent free coefficients. Retain

$$\theta_{kj} - \theta_{k,j-1} \mid \lambda_{kj}^\theta, \zeta_k \sim \mathcal{N}(0, \lambda_{kj}^\theta / \zeta_k),$$

with proper initial-coefficient, local-multiplier, and global-precision priors. The current bounded exponential/local and Gamma/global families are possible starting choices. Any fixed coefficient must be justified by basis support; do not copy an amyloid coefficient index to tau automatically.

Scale identification. Omitting speed intercepts separates the mean individual speed from the shared rate scale. Centering $s(c_A) = 0$ separates a constant coupling multiplier from the tau rate scale. The one-coefficient coupling has no redundant intercept. If a coupling spline replaces it, remove constant/null directions after centering and regularize its remaining coefficients.

TEST: constraints do not guarantee information in the data

Constant high amyloid can sustain rapid tau progression, but its multiplier can resemble a subject's residual tau speed. Similar amyloid and tau levels across subjects can make their rate functions hard to separate. Sparse data also allow earlier departure to trade against faster growth. Test recovery of β, ρ_α, timing, and speeds jointly; inspect prior sensitivity even when MCMC diagnostics look good.

10 How the sampler handles all components jointly

Use (22) with state indicators marginalized. A Metropolis-within-Gibbs update proposes one coordinate block while holding other sampled coordinates fixed; its acceptance ratio includes every changed posterior factor and any proposal-density correction. Exact slice or Gibbs steps can coexist with Metropolis steps. No particular blocking is assumed optimal.

1. **Build population functions.** Compute both rate clocks and measurement functions, including defined tails. Build the coupling function from β and fixed d_A^{coup}. Cache unaffected functions.
2. **Build each candidate amyloid path and exposure.** Obtain α_{Ai} from (18); evaluate A_i^1 at amyloid scan ages and age-quadrature nodes. Integrate m_i to construct C_i. Include the amyloid departure breakpoint in the quadrature partition.
3. **Evaluate tau and the transformed prior.** Invert C_i for α_{Ti}, check support, and evaluate (21) at tau scan ages. Compute both Weibull densities, the single timing-copula correction, and both Jacobians.
4. **Evaluate three branches.** Compute observation densities and the three weighted log-branches, then combine by log-sum-exp. Do not condition on a hard class assignment.
5. **Update subject histories.** Compare paired blocks $(\widetilde{x}_{Ai}, z_{Ai})$, $(\widetilde{x}_{Ti}, z_{Ti})$ with a four-coordinate block. Update flat levels separately initially. Compare effective samples per unit time for scientific targets, not acceptance rate alone.
6. **Update population quantities.** Use suitable slice/Metropolis blocks for rates, noise, regressions, scales, ρ_α, β, and susceptibility. Recompute all changed descendants, including coordinate-prior factors.

A consequential parameterization detail. At fixed $\widetilde{x}_{Ti}$ and z_{Ti}, an amyloid-path or β update changes C_i and therefore generally changes the *implied* α_{Ti}. Its acceptance calculation includes the changed tau departure density, timing copula, and Jacobian, as well as tau observations. A statement that "tau departure stays fixed during an amyloid update" is true only when departure age itself is held fixed as a sampled coordinate.

Two legitimate implementations. Direct sampling of $(\alpha_{Ai}, \alpha_{Ti}, z_{Ai}, z_{Ti})$ targets (16) and needs no location Jacobians. Sampling locations targets (22). Their posterior predictions should agree. Direct-age sampling is a useful small-data reference implementation, even if it mixes less efficiently.

Three distinct independence statements. Measurement errors are conditionally independent given histories and state. Residual speeds are independent in the population model conditional on covariates and Θ. The timing pair is independent of the speed pair in that same prior model. None of these statements implies posterior independence or marginal independence of amyloid and tau records.

TEST: complete dependency tracking

Amyloid rate, launch, timing, and speed updates can change tau predictions. Coupling updates can change transformed-prior terms. Global updates may affect all subjects, including auxiliary coordinates in apparently inactive classes. Check cached versus fully recomputed log-posteriors. Never omit a factor merely because it looks like a prior rather than an observation likelihood.

11 Implementation changes and mathematical verification

The supplied R files implement an amyloid-only model. The changes below are a development plan, not completed edits to those files.

Component	Required revision
`00_settings.R`	Add tau-specific settings, d_A^{coup}, coupling/correlation priors, numerical tolerances, and explicit tail conventions.
`01_integrator.R`	Retain biomarker clocks; add subject age quadrature, monotone exposure inversion, and consistent derivatives/tails.
`02_bases.R`	Build separate rate/noise bases with defined extensions. Initial coupling uses a fixed sigmoid, not a new flexible spline block.
`03_densities.R`	Add joint timing/location density with the new tau Jacobian; replace two branches by (15). Remove any proposed speed-copula correction.
`04_model.R`	Add tau quantities, q_i, ρ_α, β, and all dependencies. Keep z_{Ai}, z_{Ti} independent.
Runner and figures	Configure and compare blocks; retain chain identities; report exposure/coupling, states, conditional histories, and diagnostics.

REQUIRED: deterministic and density checks before fitting

1. **Uncoupled limit.** At $\beta = 0$, $m_i = 1$, $C_i(a) = a - \widetilde{a}_{Ti}$, the old tau age map and Jacobian are recovered, and both trajectories reduce to their separate-clock forms. At $\rho_\alpha = 0$, the timing copula density equals one.
2. **Constant-exposure limit.** If amyloid stays at a fixed level giving multiplier $m_0 > 0$, then $C_i(a) = m_0(a - \widetilde{a}_{Ti})$ and $J_{Ti} = e^{-\delta_{Ti}} D_T(\widetilde{x}_{Ti})/m_0$. High stable amyloid must still affect tau speed.
3. **Round trips and derivatives.** Map ages to locations and back, including future departures. Compare the analytic/implemented Jacobian determinant with finite differences. Check the nonzero amyloid-to-tau cross-derivative.
4. **ODE agreement.** Compare the clock solution with an independent high-accuracy ODE solve. Include departure breakpoints, asynchronous scans, slow rates, and steep coupling transitions.
5. **Probability checks.** Verify Weibull and joint copula normalization and marginal recovery; transformed densities integrate to one. Verify branch weights and posterior state probabilities sum to one.
6. **Marginalization checks.** With no tau scans, integrate unused tau coordinates and recover the amyloid-only mixture. With no data, recover the specified population and latent priors, including the β and ρ_α priors.

Use stable log-CDF/survival and `log1p`/`expm1` computations near tails. Mathematical age bounds are genuine support constraints; lookup-table overflow is a numerical issue to resolve. If intentional extra truncation is introduced, derive its normalizer under the coupled joint distribution rather than retaining the old density unchanged.

12 Empirical tests, interpretation, and reporting

REQUIRED: model assessment before substantive conclusions

1. **Simulation recovery.** Generate full latent histories and noisy scans using the actual age schedules, missingness patterns, and tau sample sizes. Vary zero/nonzero coupling, weak/strong timing correlation, departure gaps, flat high amyloid, and cure fractions. Examine interval coverage, bias, and state classification, not only trajectory fit. Include denser follow-up to distinguish design limitations from implementation errors.

2. **Comparable model fits.** Use identical marginal priors and measurement models to compare: (a) no timing dependence/no coupling; (b) timing copula only; (c) coupling only; and (d) timing copula plus coupling. The former timing-and-speed-copula model can be an additional benchmark. Do not presume the combined model wins.

3. **Prediction.** Hold out later tau scans to assess forecasting, conditioning only on information available at the forecast date. Separately hold out whole subjects for population prediction. Check predictive calibration by amyloid burden, tau burden, and follow-up duration; avoid leaking future amyloid scans into a claimed prospective forecast.

4. **Sensitivity.** Vary the coupling width/shape/prior, timing-correlation prior, departure bounds, tau susceptibility prior, rate smoothing, noise model, and tail continuation. Assess the three-state restriction and informative missingness. Test flexible coupling or stage dependence only when data support the added parameters.

5. **MCMC.** Report rank-normalized split-$\widehat{R}$, bulk/tail ESS, Monte Carlo error, and traces for population quantities, state probabilities, and representative conditional histories. Use meaningfully dispersed feasible starts. Good mixing does not establish identifiability or rule out missed modes.

Scientific interpretation. The timing copula describes related departure ages. The multiplier describes an amyloid-burden association with tau rate conditional on individual speed and tau burden. Both mechanisms can coexist, but neither establishes a causal intervention effect from observational PET alone. Correlation does not enforce ordered departures; if ordering is later required, change the joint age distribution and its normalizer explicitly.

Evidence for amyloid-associated cortical tau spread motivates testing a coupling model [1]. Stage-specific contributions of amyloid and existing tau motivate sensitivity to the separable rate assumption [2]. Tau increases without detectable amyloid accumulation, particularly in medial temporal regions, make the three-state assumption region-dependent [3]. These findings motivate model choices; they do not validate the proposed sigmoid, lifetime classes, or treatment interpretation.

Reporting boundary. Report departure ages conditional on susceptibility, with uncertainty; distinguish them from observed threshold crossings. Lifetime non-accumulation is model-based extrapolation. Before fitting, specify the tau region/tracer, calibration, age bounds, coupling width, and prior hyperparameters.

References

[1] Sanchez JS et al. The cortical origin and initial spread of medial temporal tauopathy in Alzheimer's disease assessed with positron emission tomography. *Science Translational Medicine*. 2021;13:eabc0655. Full text.

[2] Cai Y et al. Initial levels of β-amyloid and tau deposition have distinct effects on longitudinal tau accumulation in Alzheimer's disease. *Alzheimer's Research & Therapy*. 2023. doi:10.1186/s13195-023-01178-w.

[3] Costoya-Sánchez A et al. Increased medial temporal tau positron emission tomography uptake in the absence of amyloid-β positivity. *JAMA Neurology*. 2023;80:1051–1061. doi:10.1001/jamaneurol.2023.2560.